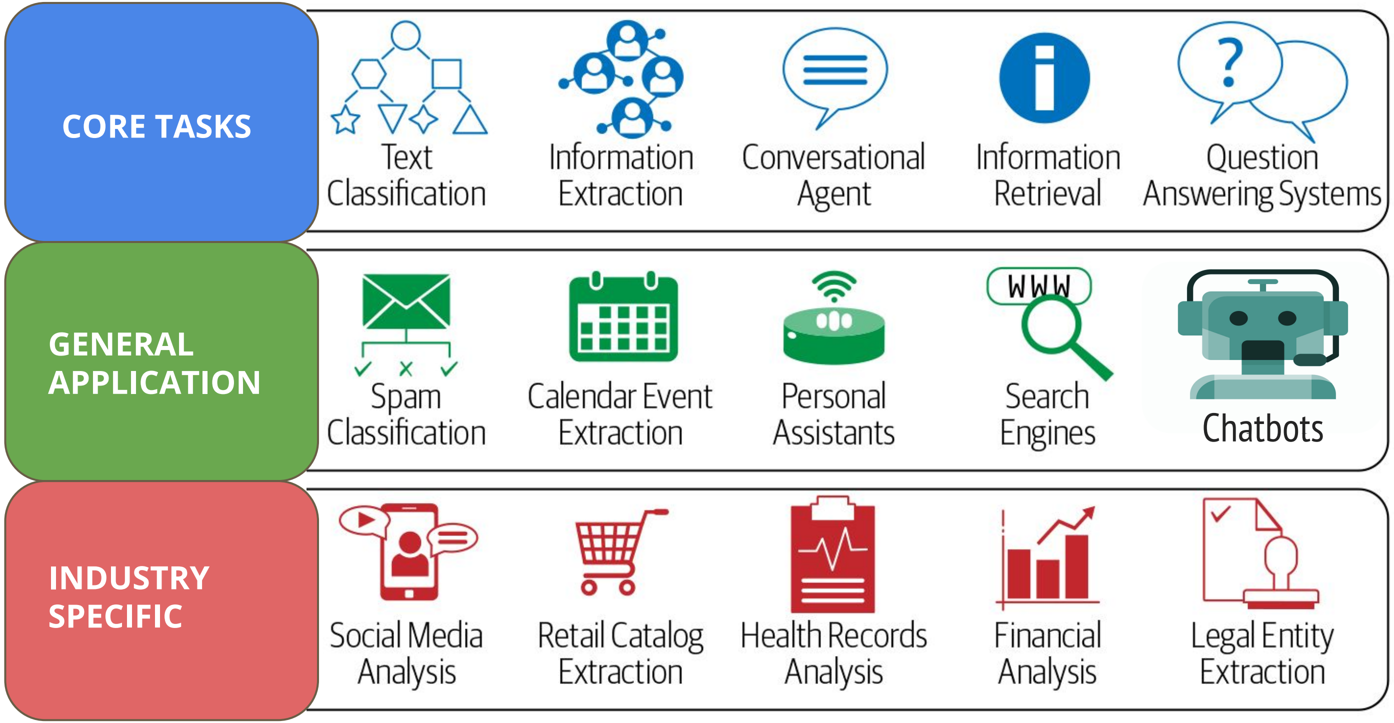
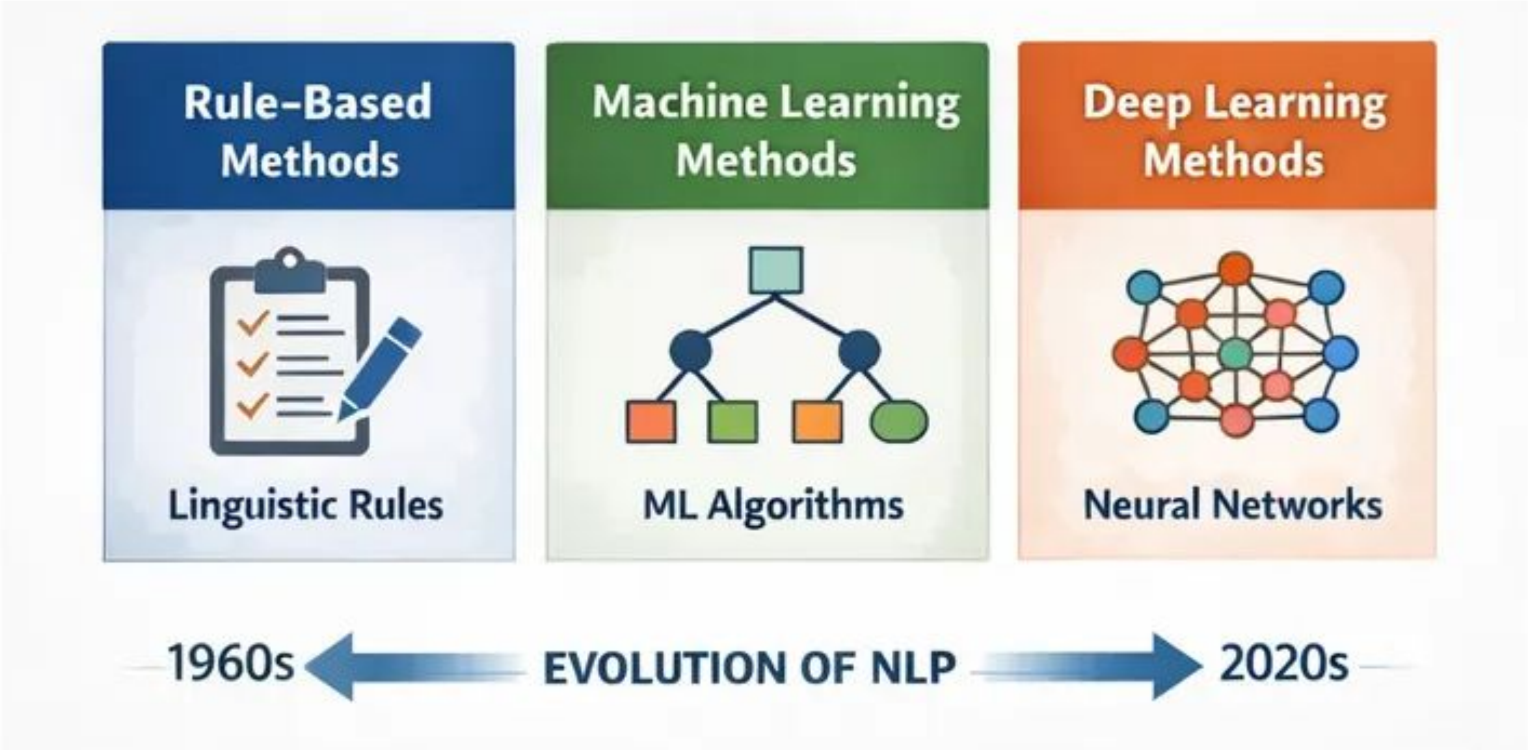
The Basic of NLP Techniques

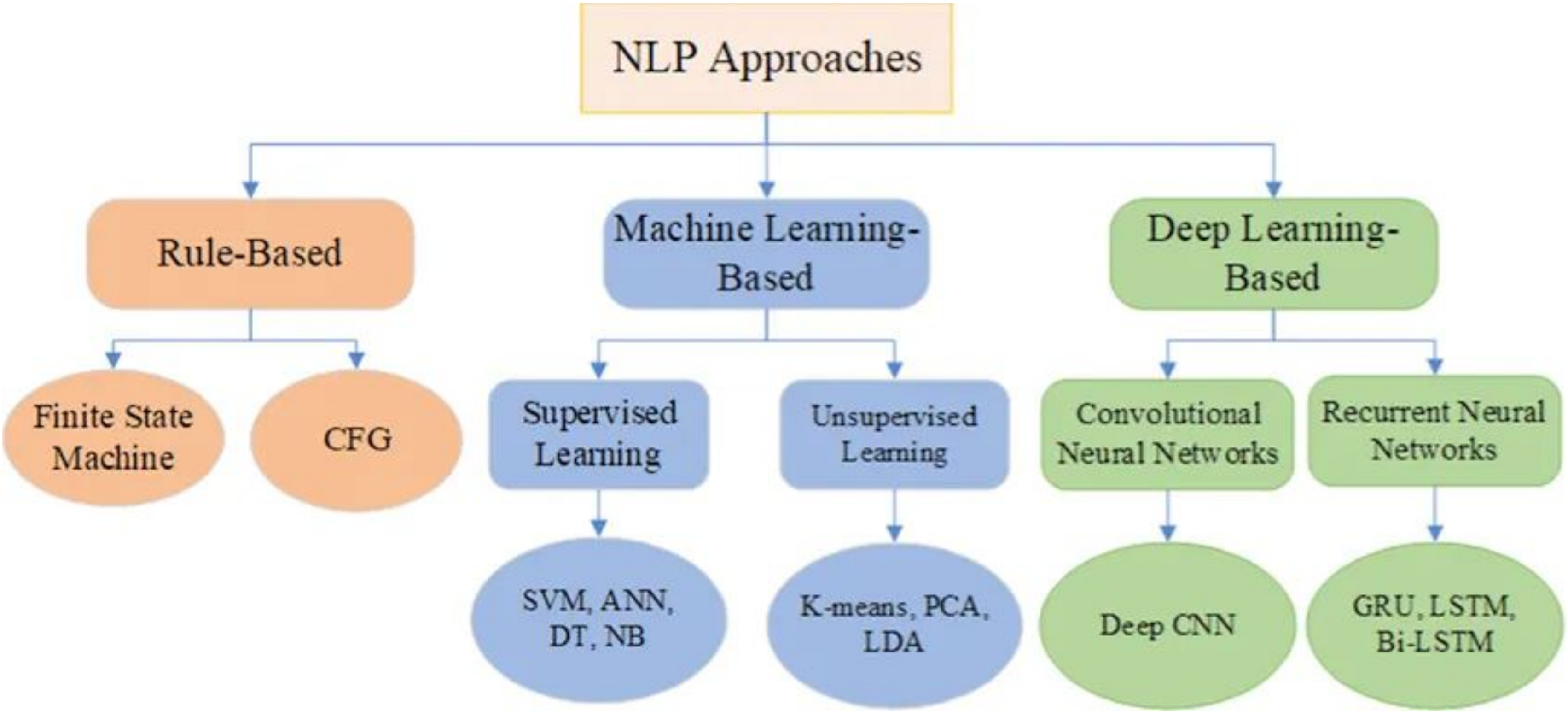
Week 3 - NLP Class

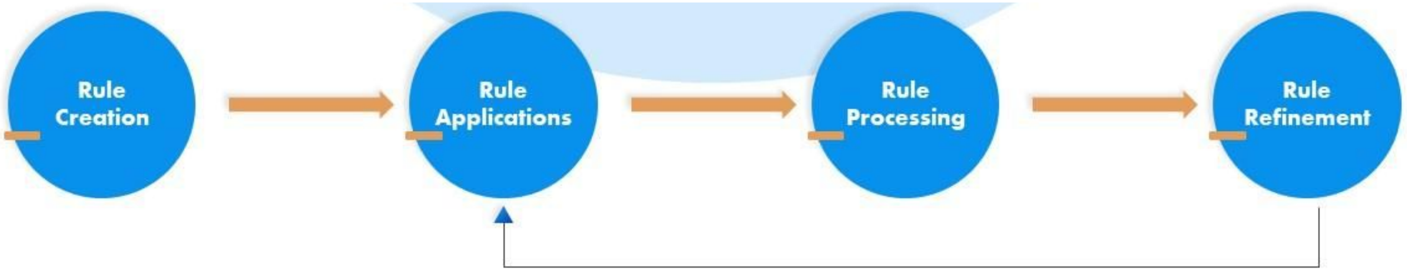
NLP Tasks and Its Applications



NLP Approaches




NLP Approaches
Rule-Based
Machine Learning-Based
Deep Learning-Based
Finite State Machine
CFG
Supervised Learning
Unsupervised Learning
Convolutional Neural Networks
Recurrent Neural Networks
SVM, ANN, DT, NB
K-means, PCA, LDA
Deep CNN
GRU, LSTM, Bi-LSTM



Step 1

Rule Creation

Based on desired tasks, domain-oriented linguistic rules are developed such as grammar rules, syntax patterns, semantic rules, etc.

Step 2

Rule Application

Predefined rules are deployed to data fed to cater matched patterns

Step 3

Rule Processing

Textual data is analyzed in context with results of matched rules for information extraction or make decisions

Step 4

Rule Refinement

Rules are developed iteratively refined by repetitive processing to enhance performance and accuracy. As per prior feedback, rules are altered or updated

NLP Approaches

- **Dictionary-based:** involves the use of a predefined list of words (a dictionary). The algorithm compares the words in the text against this dictionary to understand their significance in text

  Example: In sentiment analysis, where words are classified as positive, negative, or neutral based on the dictionary.

- **Rule-based**: utilizes a set of predefined linguistic rules. These rules are crafted by language experts and are used to interpret or process text.

  Example: In grammar checking, rules about sentence structure and word usage guide the identification of errors. This approach relies heavily on the quality and comprehensiveness of the rules defined.

- **Knowledge-based:** built around a comprehensive framework of knowledge, often requiring complex representations of language and world knowledge. The key characteristic of knowledge-based systems is their reliance on rich, domain-specific information, which can include facts about the world, relationships between concepts, and linguistic information.

  Example: A chatbot for medical diagnosis that uses a medical knowledge base to interpret symptoms described by users and provide relevant medical advice.

NLP Approaches

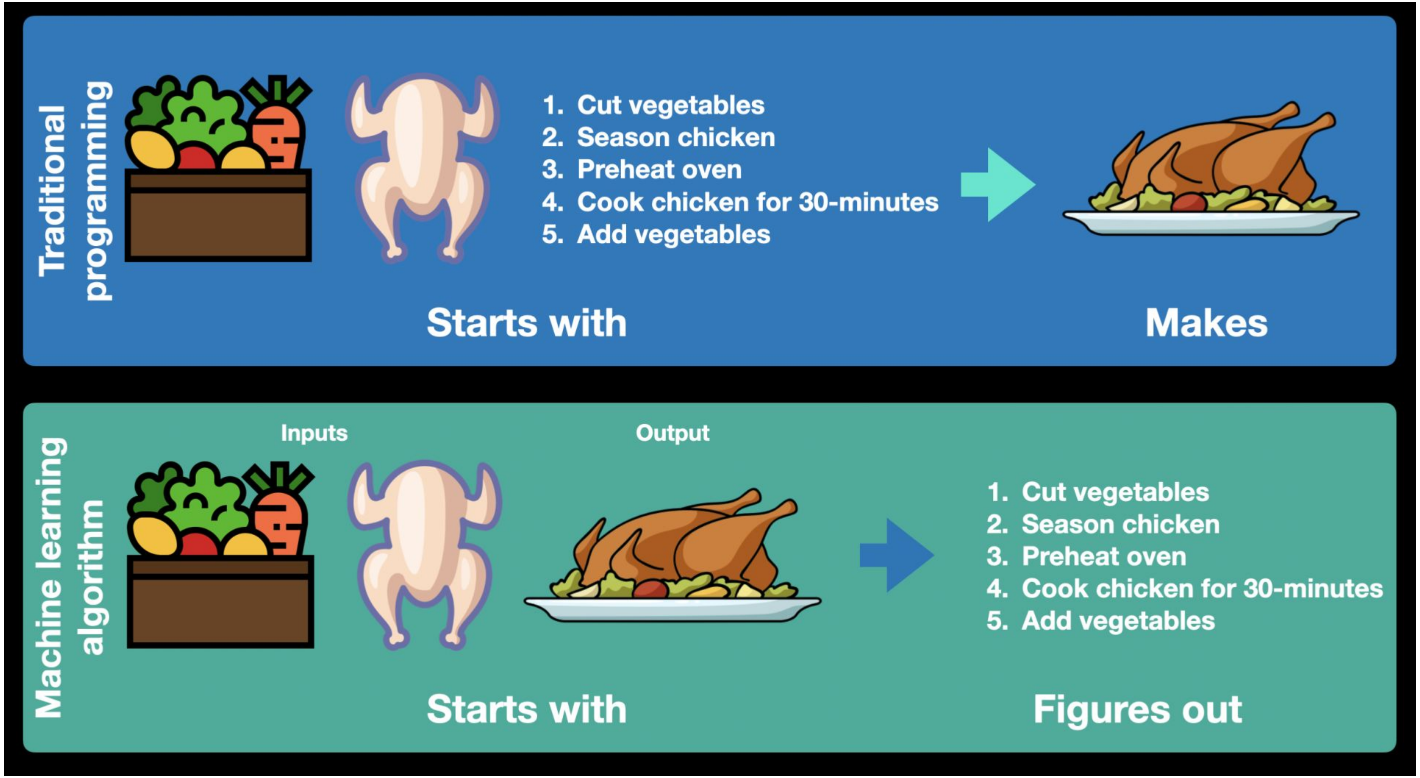
- **Machine Learning:** involves training algorithms on large datasets to learn from examples. In NLP, machine learning models (like decision trees, support vector machines, or ensemble methods) are trained on text data to perform tasks like classification, translation, or sentiment analysis. These models learn patterns and features from the training data, enabling them to make predictions on new, unseen data.

  Example: Email spam filters that use machine learning to learn from examples of labelled data such as spam and non-spam labels to effectively classify incoming emails.

- **Hybrid:** Hybrid approaches combine rule-based or dictionary-based methods with machine learning techniques to leverage the strengths of both approaches.

  Example: a hybrid system might use machine learning for broad analysis and rule-based methods for specific linguistic features.

Traditional Programming vs. ML Algorithm



NLP Approaches

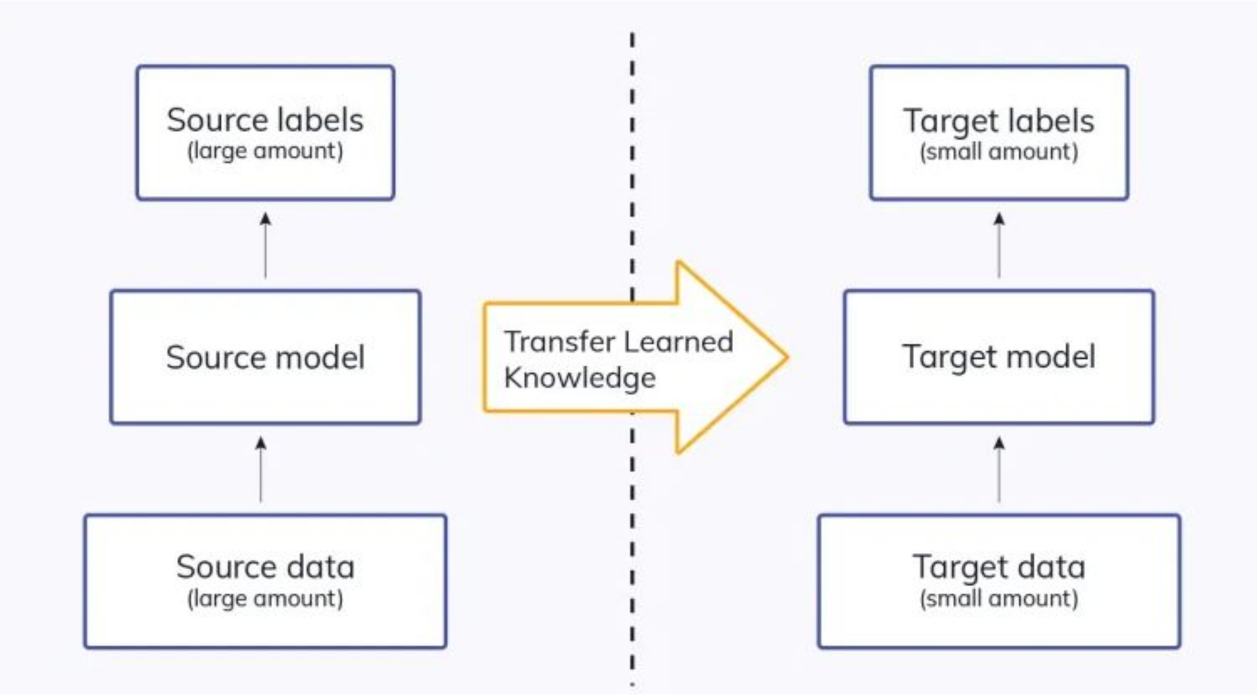
- **Neural Network/Deep Learning:** uses neural networks, particularly deep learning architectures like Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), for processing and learning from textual data. These models are capable of learning complex patterns and representations of language.

  Example: Machine translation, and speech recognition.

- **Transfer Learning:** Involves taking a pre-trained model (usually on a large, generic dataset) and fine-tuning it on a smaller, specific dataset for a particular task. This approach leverages the general language understanding of the pre-trained model, making it efficient for tasks with limited training data.

  Example: Fine-tuning a pre-trained language model like BERT (Bidirectional Encoder Representations from Transformers) for a specific task like sentiment analysis on product reviews.

Transfer Learning



Pros & Cons

Dictionary-based	Rule-based	ML	NN/DL	Transfer Learning